

GRADIENTAG MODULI SPACES QUOTIENTS

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1. ORBITS AND STABILISERS

To recall the bigger picture: We want to think of moduli spaces as geometric solutions to a classification problem, and the classical construction is as follows:

- (1) *(Over)Parametrise*: describe all objects using a parameter space X
 - (2) *Describe symmetries*: encode symmetries + extraneous data with a group action G on X whose orbits become the equivalence one is interested in
 - (3) *Quotient*: Parametrise the closed orbits by glueing orbits whose closures meet
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To recap where we left off:

Definition. Let G be an affine algebraic group, X a scheme, with $\sigma : G \times X \rightarrow X$ the G -action and $x \in X$ be a k -point.

- The *orbit* $G \cdot x$ of x is $G \cdot x = \{ g \cdot x : g \in G \} = \{ \sigma(-, x) : G(k) \rightarrow X(k) \}$
- The *stabiliser* G_x of x is $G_x = G \times_X \text{Spec}(k)$, the fibre product

$$\begin{array}{ccc}
 G_x := G \times_X \text{Spec}(k) & \longrightarrow & G \\
 \downarrow & & \downarrow \sigma_x \\
 \text{Spec}(k) & \xrightarrow{x} & X
 \end{array}$$

We say that G -action is *closed* if all G -orbits in X are closed.

- (Aside) G_x is a closed subscheme of G and is also a subgroup of G (since base-change of closed embedding is closed embedding; G_x is a subgroup since fibre products exist in AffGrp).

Example 1. Let $\mathbb{G}_m = \text{Spec}(\mathbb{k}[t, t^{-1}])$ act on $\mathbb{A}^2$ by $t(x, y) = (tx, t^{-1}y)$. The orbits are:

- origin (this is closed)
- punctured axes; this is not closed (origin is in closure)
- $\{(x, y) : xy = a\}$ for $a \in \mathbb{A}^1 \setminus \{0\}$ (these are closed)

Example 2. Let $\mathbb{G}_m$ act on $\mathbb{A}^n$ by $t(a_1, \dots, a_n) = (ta_1, \dots, ta_n)$. The orbits are:

- origin (this is closed)
- punctured lines through origin (not closed)

Lemma 1 (Lemma 3.22). Let G be an affine algebraic group acting on a scheme X over $\mathbb{k}$.

- (i) If G is an affine group variety, and $Y, Z \subseteq X$ are subschemes with Z -closed, then $\{g \in G : gY \subset Z\}$ is closed.
- (ii) If X is a variety, then for any subgroup $H \subset G$, the collection of fixed points

$$X^H = \{x \in X : Hx = x\}$$

is closed in X .

Proof. See main notes (Lemma 3.22). □

Lemma 2. Let G be an affine algebraic group acting on a scheme X . For any $x \in X(\mathbb{k})$, we have

$$\dim(G) = \dim(G_x) + \dim(G \cdot x)$$

Proof. See main notes for proof (Proposition 3.21). □

2. QUOTIENT

Suppose we have a scheme X with action given by G . Can we form a quotient X/G ?

- if $X \subseteq \mathbb{A}^n$ is a variety, and we want to start identifying points in $\mathbb{A}^n$, how to embed this quotient into $\mathbb{A}^n$?
- Better idea: look at the coordinate rings since functions on X/G (should) correspond to functions on X invariant under $G \curvearrowright$ we look at the invariant ring, e.g. $(\mathbb{k}[x_1, \dots, x_n]/I(X))^G$ for some algebraic set $X \subset \mathbb{A}^n$.

Before introducing a *good quotient*, we must introduce some notation.

Let G be an affine algebraic group, $(X, \mathcal{O}_X)$ be a scheme. Then G acts on the $\mathbb{k}$ -algebra $\mathcal{O}_X(X)$ by

$$g \cdot f(x) = f(g^{-1}x).$$

A morphism $\phi : X \rightarrow Z$ is *G -invariant* if for all $g \in G$ we have $g \cdot \phi(x) = \phi(g^{-1}x) = \phi(x)$, i.e. the diagram commutes:

$$\begin{array}{ccc} G \times X & \xrightarrow{g} & X \\ \downarrow \pi_2 & & \downarrow \phi \\ X & \xrightarrow{\phi} & Z \end{array}$$

That is, ϕ satisfies the equation $\phi \circ \pi_2 = \phi \circ \sigma$ where π_2 is projection onto 2nd coordinate, σ is the G -action.

We denote the k -subalgebra of G -invariant functions to be

$$\mathcal{O}(X)^G = \mathcal{O}_X(X)^G = \{ f \in \mathcal{O}(X) : g \cdot f = f, \forall g \in G \}.$$

For any $U \subset X$, we say that U is G -invariant subset if $GU \subset U$. Whenever U is G -invariant, then G acts on the k -subalgebra $\mathcal{O}_X(U)$ and we denote $\mathcal{O}_X(U)^G$ to be the subalgebra of invariants.

Definition. Let X be a scheme with an affine group G acting on X . A *good quotient* (Y, φ) is a morphism $\varphi : X \rightarrow Y$ of schemes such that

- (1) φ is G -invariant
- (2) φ is surjective
- (3) If $U \subset Y$ is open, then the morphism $\mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(\varphi^{-1}(U))$ is an isomorphism onto $\mathcal{O}_X(\varphi^{-1}(U))^G$
- (4) If $W \subseteq X$ is a G -invariant closed subset of X , then $\varphi(W)$ is closed in Y
- (5) If $W, W' \subseteq X$ are disjoint G -invariant closed, then $\varphi(W)$ and $\varphi(W')$ are disjoint
- (6) φ is affine (i.e. preimage of every affine open is affine)

If the preimage of each point is a single orbit, then (Y, φ) is a *geometric quotient*.

Example 1. (Example 3.31) We return to Example 1 where $\mathbb{G}_m$ acting on $\mathbb{A}^2$ by $t(x, y) = (tx, t^{-1}y)$. The orbits $\{ (x, y) : xy = a \} = H_a$ for $a \in \mathbb{A}^1 \setminus \{0\}$ are closed and are parametrised by a . The remaining orbits contain the origin in their closures, and so we suspect that we can parametrise the 3 orbits will be identified together in the quotient.

Claim: $(\mathbb{A}^1, (x, y) \mapsto xy)$ is a *good quotient* for $\mathbb{A}^2$ with the action by $\mathbb{G}_m$.

Proof. We are working with the morphism $\varphi : \mathbb{A}^2 \rightarrow \mathbb{A}^1$ that sends $(x, y) \mapsto xy$.

- (1) φ is G -invariant since

$$\varphi(t(x, y)) = \varphi(tx, t^{-1}y) = xy = \varphi(x, y).$$

- (2) φ is surjective since any $z \in \mathbb{A}^1$, we have $(z, 1) \mapsto z \cdot 1 = z$.
- (3) Let $U \subseteq \mathbb{A}^1$ be open and consider the morphism

$$\varphi^* : \mathcal{O}_{\mathbb{A}^1}(U) \rightarrow \mathcal{O}_{\mathbb{A}^2}(\varphi^{-1}(U))$$

For $\mathbb{A}^1$, all closed sets are $\emptyset, \mathbb{A}^1$ and finite collection of closed points and so without loss of generality, suppose $U = \mathbb{A}^1$ or $U = \mathbb{A}^1 \setminus \{a_1, \dots, a_r\}$.

In the case where $U = \mathbb{A}^1$, let $t \in \mathbb{G}_m$ act on $\mathcal{O}(\mathbb{A}^2) = \mathbb{C}[x, y]$ which gives us

$$t(\sum_{i,j} a_{ij} x^i y^j) = \sum_{i,j} a_{ij} t^{j-i} x^i y^j$$

and so the invariant subalgebra will consist of

$$\mathbb{C}[x, y]^{\mathbb{G}_m} = \{ \sum_{i,j} a_{ij} x^i y^j : a_{ij} = 0, \text{ whenever } i \neq j \} = \mathbb{C}[xy]$$

In the latter case, where $U = \mathbb{A}^1 \setminus \{a_1, \dots, a_r\}$ is an open we get that

$$\begin{aligned} \mathcal{O}_{\mathbb{A}^1}(U) &= \mathbb{C}[z]_f & \text{where } f(z) &= \prod (z - a_i) \\ \mathcal{O}_{\mathbb{A}^2}(\varphi^{-1}(U)) &= \mathbb{C}[x, y]_F & \text{where } F(x, y) &= f(xy) \end{aligned}$$

and so

$$\mathcal{O}_{\mathbb{A}^2}(\varphi^{-1}(U))^{\mathbb{G}_m} = (\mathbb{C}[x, y]_F)^{\mathbb{G}_m} = (\mathbb{C}[x, y]^{\mathbb{G}_m})_F = \mathbb{C}[xy]_F = \mathcal{O}_{\mathbb{A}^1}(U).$$

For (4) and (5) the following lemma (proof included at the end) helps clean up notation:

Lemma 3. With $\varphi : \mathbb{A}^2 \rightarrow \mathbb{A}^1$ the map $(x, y) \mapsto xy = z$, equip a grading on $\mathbb{k}[x, y]$ with $\deg(x) = 1$ and $\deg(y) = -1$. Let $I \subset \mathbb{k}[x, y]$ be an ideal and let $I_0 := I \cap \mathbb{k}[xy] = I \cap \mathbb{k}[z]$ be ideal with degree 0 grading. Then $\varphi(W) = V(I_0)$ for some closed G -invariant $W \subset \mathbb{A}^2$.

Proof of Lemma. Suppose $a \in \varphi(W)$ for some closed G -invariant $W \subset \mathbb{A}^2$. This gives us that there exists some $(x, y) \in \mathbb{A}^2$ such that $xy = a$ and so any $f \in I_0$ gives $f(a) = f(xy) = 0$ and so $a \in V(I_0)$.

Conversely, let $a \in V(I_0)$, and suppose for the sake of contradiction that $a \notin \varphi(W)$ for some G -invariant closed $W \subset \mathbb{A}^2$. It follows that

$$W \cap V(xy - a) = \emptyset \iff I + (xy - a) = (1),$$

and so there exists some $f \in I$ and $g \in \mathbb{k}[x, y]$ such that

$$f + (xy - a)g = 1.$$

If $a = 0$, we are left with $f + (xy)g = 1$. Expressing $f = \sum_i f_i$ and $g = \sum_i g_i$ where f_i, g_i denote the degree i components, by looking at the degree 0 components, we are left with

$$f_0 + xy \cdot g_0 = 1$$

but $f_0 \in I_0$, and so we have that $1 \in I_0 + (xy)$. Since we also have that $0 \in V(I_0)$, we get $I_0 \subset (xy)$ and so $I_0 + (xy) \subset (xy)$, giving us that $1 \in (xy)$, a contradiction.

In the case where $a \neq 0$, we have

$$f(x, y) + (xy - a) \cdot g(x, y) = 1.$$

Evaluating $x = t$ and $y = at^{-1}$, we get for each degree i component that

$$f_i(t, at^{-1}) = t^i \cdot f_i(1, a)$$

which gives us

$$f = \sum_i f_i = \sum_i t^i \cdot f_i(1, a) = 1.$$

Thus, for any $i \neq 0$, we conclude that $f_i = 0$ and $f_0(1, a) = 1 \in I_0$. Since $a \in V(I_0)$, we get that $f_0(a) = 0$, but $f_0(a) = f_0(xy)$ with $xy = a$. Thus, we have $0 = f_0(a) = f_0(1, a) = 1$, giving us a contradiction. Thus, $a \in \varphi(W)$. $\square$

- (4) Let $W \subset X$ be a G -invariant closed subset of X .

Let $W \subseteq X$ be closed, G -invariant. It follows that $W = V(I)$ for some ideal I , and by Lemma 3, we get that $\varphi(W) = V(I_0)$ where $I_0 = I \cap k[x, y]$. Thus, $\varphi(W)$ is closed.

- (5) Let W_1, W_2 be disjoint G -invariant closed subsets.

Since W_1, W_2 are closed, we can write them as

$$V(I) = W_1, V(J) = W_2 \implies V(I + J) = V(I) \cap V(J) = \emptyset$$

and so we get that $I + J = (1)$. Looking at the degree 0 components, we are left with

$$I_0 + J_0 = (1)$$

- (6) φ is affine since it is a morphism of affine schemes.

$\square$

The example above is **not** a geometric quotient, since $\varphi^{-1}(0)$ is the union of 3 orbits.

Corollary 4. Let G be an affine algebraic group acting on X , and $\varphi : X \rightarrow Y$ be a good quotient.

- (a) $\varphi(x_1) = \varphi(x_2) \iff \overline{G \cdot x_1} \cap \overline{G \cdot x_2} \neq \emptyset$
- (b) For each $y \in Y$, $\varphi^{-1}(y)$ contains a unique closed orbit. Moreover, if the group action is closed, then φ is a *geometric quotient*.

Proof. (a) Suppose $\varphi(x_1) = \varphi(x_2)$, and since these are not disjoint, we get that the orbits $Gx_1 \cap Gx_2 \neq \emptyset$. Conversely, suppose the closure of orbits are nonempty. Since φ is constant on orbit closures, we get our result.

(b) Suppose we had two closed orbits W_1, W_2 in $\varphi^{-1}(y)$. Since their images are $\varphi(W_i) = y$, we contradict the disjointness of closed disjoint G -invariant subsets of good quotients. $\square$

Proposition 5. (1) If $\varphi : X \rightarrow Y$ is a good (resp. geometric) quotient, then after restricting to any open $U \subset Y$, the restriction

$$\varphi|_{\varphi^{-1}(U)} : \varphi^{-1}(U) \rightarrow U$$

is a good (resp. geometric) quotient of G acting on $\varphi^{-1}(U)$.

- (2) If $\varphi : X \rightarrow Y$ is a G -invariant morphism, and we have open cover $\{U_\alpha\}$ of Y such that each $\varphi_\alpha := \varphi|_{\varphi^{-1}(U_\alpha)} : \varphi^{-1}(U_\alpha) \rightarrow U_\alpha$ are all good (resp. geometric) quotients, then so is $\varphi : X \rightarrow Y$.

Proof. Exercise :-)

$\square$

Definition. Let X be a scheme and G be an affine algebraic group acting on X . A *categorical quotient* (Y, φ) is a G -invariant morphism of schemes $\varphi : X \rightarrow Y$ satisfying the universal property: for any G -invariant morphism of schemes $f : X \rightarrow Z$, there exists a unique morphism $\tilde{f} : Y \rightarrow Z$ such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & Z \\ & \searrow \varphi & \nearrow \exists! \tilde{f} \\ & Y & \end{array}$$

- if for all k -points $y \in Y$, we have $\varphi^{-1}(y)$ consists of a single orbit, then we say $Y = X//G$ is the *orbit space*
- A categorical quotient is orbit space if the G -action is closed on X (since φ is constant on orbits, and thus on orbit closures)

Of course, since (Y, φ) are universal, it is unique (up to unique isomorphism) if it exists.

Remark. If $\varphi : X \rightarrow Y$ is a G -invariant morphism, and we have open cover $\{U_\alpha\}$ of Y such that each $\varphi_\alpha := \varphi|_{\varphi^{-1}(U_\alpha)} : \varphi^{-1}(U_\alpha) \rightarrow U_\alpha$ are all categorical quotients, then so is $\varphi : X \rightarrow Y$.

Proposition 6. Let $\varphi : X \rightarrow Y$ be a categorical quotient with a G -action on X .

- (1) If X is connected, then Y is connected.
- (2) If X is irreducible, then Y is irreducible.
- (3) If X is reduced, then Y is reduced.

Proof. Exercise :-)

□

Helpful lemma for Proposition 6:

Lemma 7. $\varphi : X \rightarrow Y$ is a categorical quotient with a G -action on X . Then:

- (1) φ is an epimorphism
- (2) φ is dominant.

Proof. (1) Let $u, v : Y \rightarrow Z$ be morphisms such that $\underbrace{u \circ \varphi}_{\text{call } f} = v \circ \varphi : X \rightarrow Z$. Since φ is

G -invariant, it follows that if $\sigma : G \times X \rightarrow X$ denotes the G -action, then

$$\begin{aligned} u \circ (\varphi \circ \sigma) &= v \circ (\varphi \circ \pi_2) \implies f \circ \sigma = f \circ \pi_2 \\ &\implies f \text{ is } G\text{-invariant too} \\ &\implies \exists! h : Y \rightarrow Z \text{ such that } h \circ \varphi = f \quad \text{by universal property} \\ &\implies u = v = h. \end{aligned}$$

- (2) Consider the closure $\overline{\varphi(X)} \subset Y$ and let $\iota : \overline{\varphi(X)} \rightarrow Y$ be the canonical inclusion. Since $\varphi(X) \subset \overline{\varphi(X)}$, there exists a morphism $f : X \rightarrow \overline{\varphi(X)}$ such that $\varphi = \iota \circ f$. To help

keep track, consider the (not-yet commutative) diagram:

$$\begin{array}{ccc} X & \xrightarrow{f} & \overline{\varphi(X)} \\ & \searrow \varphi & \swarrow \iota \\ & Y & \end{array} \quad \begin{array}{c} \nearrow \exists! h \\ \text{dashed arrow} \end{array}$$

Claim: f is G -invariant.

Since φ is G -invariant, we get that

$$\begin{aligned} \varphi \circ \sigma = \varphi \circ \pi_2 &\implies \iota \circ f \circ \sigma = \iota \circ f \circ \pi_2 \\ &\implies f \circ \sigma = f \circ \pi_2 \quad \text{since } \iota \text{ is closed immersion } \implies \text{monomorphism} \end{aligned}$$

By the universal property, there exists a unique morphism $h : Y \rightarrow \overline{\varphi(X)}$ such that $f = h \circ \varphi$. Thus, we have that

$$\varphi = \iota \circ f = \iota \circ h \circ \varphi \implies \iota \circ h \circ \varphi = \text{id}_Y \circ \varphi$$

and since φ is an epimorphism by (1), we get $\iota \circ h = \text{id}_Y$, giving us that ι has a right-inverse and since it is a closed immersion, it is an isomorphism. Thus, the image is dense and φ is dominant.

□

Example 2. From Example 2, where $\mathbb{G}_m$ acts on $\mathbb{A}^n$ by scaling, since the origin is in the closure of every orbit, any G -invariant morphism $\mathbb{A}^n \rightarrow Z$ will be a constant morphism.

Claim: The categorical quotient of this action is the map $(\text{Spec}(k), \varphi)$ where $\varphi : \mathbb{A}^n \rightarrow \text{Spec}(k)$, and we get that any other G -invariant map $f : \mathbb{A}^n \rightarrow Z$ is going to be $f = z \circ \varphi$ for some unique morphism $z : \text{Spec}(k) \rightarrow Z$.

- Follows since continuous morphisms into a separated scheme that is constant on dense open are constant on the closure

Proposition 8 (Good $\implies$ categorical, Prop 3.30). Let G be an affine algebraic group acting on a scheme X . Any (Y, φ) satisfying properties (1), (3), (4), (5) in the definition of good quotients is a categorical quotient. In particular, good quotients are categorical quotients.

Proof. We need to check that (Y, φ) is a G -invariant morphism of schemes that is universal wrt the diagram. Since (1) requires φ is G -invariant, we only need to prove the universal property.

Idea: take (finite) open cover of Z , then use this to construct a cover on Y and then locally define morphisms on local patches, then glue together to get global morphism.

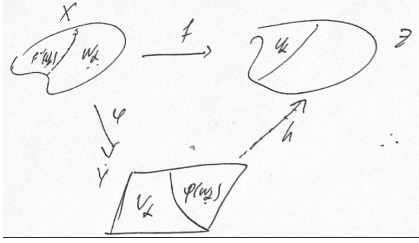
Let $\{U_\alpha\}$ be open cover of Z and suppose $f : X \rightarrow Z$ is another G -invariant morphism of schemes. Thus, it follows that

$$\begin{aligned} f^{-1}(U_\alpha) \subset X &\implies W_\alpha := X \setminus f^{-1}(U_\alpha) \text{ is closed in } X \\ &\implies \varphi(W_\alpha) \text{ is closed in } Y \quad \text{by (4) of good quotients} \end{aligned}$$

$$\implies V_\alpha := Y \setminus \varphi(W_\alpha) \text{ is open in } Y$$

$$\implies \varphi^{-1}(V_\alpha) \subset f^{-1}(U_\alpha)$$

see diagram below



Since $\{U_\alpha\}$ cover Z , we get that $\bigcap W_\alpha = \emptyset$.

Claim: We claim that $\bigcap_\alpha \varphi(W_\alpha) = \emptyset$, i.e., V_α is an open cover of Y .

Proof of Claim. Suppose for contradiction that it is nonempty. It follows that $\bigcap \varphi(W_\alpha)$ has a closed k -point. Let W be a closed orbit in the preimage of that point. Since $W \cap W_\alpha \neq \emptyset$, we get that $\varphi(W) \cap \varphi(W_\alpha) \neq \emptyset$. Now W consists of a single orbit and each W_α is G -invariant, and so we get $W \subset W_\alpha$ for each α , a contradiction. $\square$

Since f is G -invariant, we see that the homomorphism $\mathcal{O}_Z(U_\alpha) \rightarrow \mathcal{O}_X(f^{-1}(U_\alpha))^G \subset \mathcal{O}_X(f^{-1}(U_\alpha))$ and by the isomorphism by (3) of good quotients we get $\varphi^* : \mathcal{O}_Y(V_\alpha) \xrightarrow{\sim} \mathcal{O}_X(\varphi^{-1}(V_\alpha))$, we get that there must exist a unique morphism h_α^* making the diagram commute:

$$\begin{array}{ccc} \mathcal{O}_Z(U_\alpha) & \xrightarrow{\quad h_\alpha^* \quad} & \mathcal{O}_Y(V_\alpha) \\ \downarrow f^* & & \downarrow \varphi^* \\ \mathcal{O}_X(f^{-1}(U_\alpha))^G & \longrightarrow & \mathcal{O}_X(\varphi^{-1}(V_\alpha))^G \end{array}$$

Since the U_α are affine, we get that the k -algebra homomorphism h_α^* corresponds to a morphism $h_\alpha : V_\alpha \rightarrow U_\alpha$.

We see that by construction (go back to diagram)

$$f|_{\varphi^{-1}(V_\alpha)} = h_\alpha \circ \varphi|_{\varphi^{-1}(V_\alpha)} : \varphi^{-1}(V_\alpha) \rightarrow U_\alpha$$

with $h_i = h_j$ on $V_i \cap V_j$, allowing us to patch together a global morphism $Y \rightarrow Z$ such that $f = h \circ \varphi$. Uniqueness follows since h_α are all unique. $\square$